

AI Evolution as an Advancing Approximation Frontier

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Abstract

AI development is often described as progress toward increasingly capable systems. This paper asks a different question: how does successful progress change the development problems that remain? It treats AI development as an advancing approximation process in which artificial systems increasingly reproduce, exceed, or substitute for functions associated with intelligent human performance while relying on mechanisms that can differ substantially from those that generate human intelligence.

Among the important implications of this perspective, progress can change the **composition** of the unresolved development frontier rather than simply reduce its size. This motivates the empirical hypothesis that dimensions with weakly specified or difficult-to-evaluate targets may resolve more slowly than dimensions with clearer targets and feedback. Capability can also outrun evaluation, exposing behaviors that earlier evaluation regimes were not designed to detect. In addition, the reference conditions surrounding AI development need not remain fixed: AI use can alter human practices, evaluative capacity, training and feedback data, expectations, the division of cognitive labor, and increasingly the development process itself.

A further methodological implication follows. When the system being diagnosed changes while competing explanations are being tested, diagnostic learning may race against frontier movement, leaving some development attributions persistently provisional.

The paper focuses on these conditional mechanisms rather than forecasting a fixed capability ceiling or a particular architectural succession. Its central contribution is a conditional account of how AI progress can change not only how much of a development problem remains, but also the composition, evaluability, reference conditions, and developmental structure of the problems that come next.

Keywords: artificial intelligence; AI development; approximation; development frontier; evaluation lag; human–AI coevolution; prediction–attribution conjugacy

1. Introduction — Progress Changes the Problem That Remains

AI development has repeatedly changed what counts as the next difficult problem. Scaling-law work suggested that larger models, more data, and more compute could produce predictable gains. Compute-optimal training then showed that how a fixed compute budget was allocated could matter more than simply increasing parameter count. More recently, inference-time scaling has shown that how computation is used can matter as much as how much pretrained capacity a system has [1], [2], [3].

The development problem is therefore not simply the same problem becoming smaller.

This paper examines that observation as a general feature of AI development. It uses **development frontier** to mean the set and structure of unresolved problems that materially constrain or shape further AI advancement. A current **bottleneck** is one binding constraint within that frontier. What researchers choose to work on is a response to the frontier, not the frontier itself.

With that distinction in place, the paper asks:

How does continued AI progress change the composition and character of the development problems that remain?

The argument begins from a simple approximation view. Artificial systems increasingly reproduce, exceed, or substitute for functions associated with intelligent human performance through artificial mechanisms that can differ substantially from the processes that generate human intelligence. If different kinds of problems are resolved at different rates, successful progress changes the composition of what remains. New capabilities can also reveal new evaluation problems, deployment contexts, or expectations rather than merely subtracting old discrepancies.

The frontier can move on the human side as well. Widespread AI use can change what humans practice, what capabilities remain scarce or valuable, how cognitive work is distributed, and what standards are used to evaluate later systems. Meanwhile, AI increasingly contributes to AI research and engineering itself, shifting parts of the development process from human-only production toward mixed human–AI production.

This paper’s contribution is a conditional account of how successful AI development can change the composition and character of the unresolved development frontier. It develops four mechanisms—differential resolution, evaluation lag, human-reference feedback, and AI-mediated successor development—and derives empirically examinable implications for which problems persist, when failures become visible, how development conditions change, and when frontier attributions remain identifiable. It remains agnostic about the identity of the next breakthrough, the longevity of transformer-centered development, and the eventual degree of AI autonomy in R&D. The title’s **approximation frontier** refers to the development frontier viewed through this account of advancing approximation.

The paper later returns to Prediction–Attribution Conjugacy (PAC), developed in earlier work [4], to ask a practical question: when several explanations of a frontier are plausible, which prospective observations could distinguish them before the frontier itself changes again?

2. AI Development as Approximation

Let HI_t denote the manifestations of human intelligence relevant to the domain under consideration at time t . Represent them abstractly as

$$HI_t = F_t(X_t)$$

where X_t denotes the human, social, environmental, institutional, embodied, and other conditions relevant to those manifestations, and $F_t(\cdot)$ denotes the incompletely understood process through which those conditions generate them. Here X_t is a placeholder for relevant conditions rather than a claim that they can all be enumerated.

Let AI_t denote the corresponding artificial-system capability at time t :

$$AI_t = G_t(I_t)$$

where $G(\cdot)$ is the artificial mechanism then available and I_t is its information and computational environment, including data, feedback, tools, interaction, synthetic material, and computational resources.

HI_t and AI_t denote relevant capability profiles or manifestations, not one-dimensional intelligence scores.

Here, **approximation** refers to reproducing, exceeding, or substituting for functions associated with intelligent human performance. It concerns functional performance rather than reconstruction of human cognition.

The Transformer is a useful example. Vaswani et al. replaced the recurrent and convolutional structures then dominant in sequence modeling with an architecture based on attention alone. The architecture improved translation performance while permitting greater parallelization and faster training [5]. The breakthrough came from a better artificial mechanism for performing language-related tasks rather than from a more complete theory of human language production.

A system can therefore translate better than most humans while arriving at that performance through a very different developmental and generative path. Task approximation and generative equivalence are distinct.

The distinction becomes consequential when context, social relations, institutional roles, embodied conditions, or changing norms affect what counts as an adequate answer. Those conditions may matter little for some tasks and substantially for others.

The paper’s broader residual hypothesis rests on **specifiability, evaluability, and feedback availability**. Those difficulties can arise even when an artificial system reproduces more of the human generating process, because humans themselves may disagree about what counts as an adequate outcome:

Residual dimensions that are difficult to specify or evaluate may be resolved more slowly than dimensions with clear targets and feedback.

The F/G distinction highlights one important sub-case: observable outputs may be easier to approximate than the contextual or evaluative conditions that determine when those outputs are adequate.

3. Differential Resolution Changes the Residual

Bottleneck migration is a familiar feature of technological development. Histories of large technical systems describe “reverse salients”: lagging components whose resolution exposes the next critical problem in an advancing system [6]. AI development provides a contemporary version of that broader pattern.

Scaling offers a clear sequence. Early scaling-law work showed regular relationships among model size, data, compute, and loss [1]. Compute-optimal training then changed the practical diagnosis: under a fixed training budget, inefficient allocation between parameters and training tokens could be more limiting than raw scale. Chinchilla operationalized that revised diagnosis and outperformed substantially larger models under the same training-compute budget [2].

Sparse mixture-of-experts methods relaxed another constraint by activating only selected experts for each input, allowing effective capacity to rise without proportional growth in computation [7]. Inference-time

scaling changed the location of productive computation again: Snell et al. showed that on some problems a smaller model using well-allocated test-time compute could outperform a much larger pretrained model under matched-compute conditions [3].

Taken together, these episodes show that **different constraints become relatively important at different stages** rather than forming a predetermined ladder from architecture to scale to allocation to inference.

The same logic can apply to the AI–human residual, but the empirical burden is higher.

By residual, this paper means the set of human-relevant discrepancies that remain materially unresolved for the comparison or task under consideration. Progress can remove some discrepancies. It can also reveal or add others as systems enter new contexts, acquire new capabilities, or face higher expectations. The residual is therefore not a subtraction-only object.

The necessary condition is simply that different dimensions resolve at different rates. That condition alone is not the substantive result. The empirical question is whether resolution rates differ systematically with properties that matter.

The paper advances a provisional hypothesis:

Well-specified and easily evaluated dimensions tend, over relevant periods, to be more tractable than dimensions whose targets are contested, contextual, weakly specified, or difficult to evaluate.

This is testable rather than definitional. Evidence against it would include broad capability improvements that reduce supposedly difficult target-side discrepancies as fast as or faster than clearer dimensions, or longitudinal comparisons showing no relationship between specifiability, evaluability, and time to reliable performance.

The distinction between **mechanism-side** and **target-side** difficulty helps make the hypothesis concrete.

A mechanism-side problem asks: how can the artificial system learn, route, store, or apply information more effectively?

A target-side problem asks: what exactly is the behavior to be approximated, and how would we know that the approximation is adequate?

“Translate this sentence into French” is relatively well specified. “Summarize this contract fairly” is different. Fairness can depend on whose interests matter, which omissions are consequential, what legal role the summary serves, and what contextual assumptions govern the judgment.

Alignment research provides more demanding versions of the same issue. A recent review of 94 influential value-alignment papers found that most did not explicitly define “values” and often relied on preferences as a proxy for more socially and culturally situated concepts [8]. Related social-choice work emphasizes that human feedback can contain genuine disagreement, making some alignment problems partly problems of aggregation rather than discovery of one latent preference [9].

Target-side difficulty can therefore arise from at least four sources: specification, plurality, context, and evaluation.

These sources of difficulty may themselves be temporary. Better architectures, richer information, improved evaluation, or a future approximation regime may reduce them substantially. The hypothesis is that some dimensions may nevertheless leave the residual more slowly than others over relevant periods.

Testing this hypothesis requires more than counting which problems appear unresolved. “Resolution” is itself observed through evaluation: hard-to-evaluate dimensions can look unresolved because measurement is weak, while laboratories may devote more effort and better benchmarks to what is already measurable. A credible empirical design would therefore need to define the comparison period in advance, use a common rule for observed resolution, and code specifiability and evaluability independently of the later outcome. Even then, observed resolution rates would reflect effort allocation, benchmark quality, and available feedback as well as underlying tractability.

A prospective test could use development problems or evaluation dimensions as units of analysis, code target specifiability, evaluability, and feedback availability before observing later outcomes, define resolution using a common threshold, and compare time-to-resolution over a fixed observation window. Effort allocation, benchmark changes, and major shifts in available data or compute would need to be recorded alongside the outcome so that observed resolution rates are not interpreted as pure measures of intrinsic difficulty.

This is a form of survivor selection: the composition of an unresolved population can change when its members leave at different rates, a mechanism developed in related work on the composition of unresolved inquiry [10]. Here, the relevant population is the set of unresolved AI-development discrepancies, and the empirical question is whether attributes such as specifiability and evaluability predict differential resolution. Progress can reduce the amount of unresolved discrepancy while changing its composition. The remainder can therefore differ in kind as well as in size.

4. Evaluation Can Lag Capability

Section 3 concerns which problems remain; this section concerns why some newly important problems may remain unseen long enough to matter.

A changing residual matters most when the remaining problems are also difficult to detect or evaluate.

Capability can advance faster than the evaluation regime designed for an earlier system. OpenAI reported in 2026 that a model operating over extended tasks exhibited failure modes during limited internal use that existing pre-deployment evaluations had not captured. The important change was that longer persistence created behavior requiring evaluation at the trajectory level rather than one action at a time [11].

This suggests a more specific mechanism than the generic observation that more capable systems can create larger consequences:

Hard-to-evaluate residual failures can persist unnoticed while capability, deployment, or autonomy expands around them.

The sequence is straightforward. A system gains a new capability or longer operating horizon. Existing evaluations continue to measure shorter or simpler behavior. A deficiency that was previously irrelevant or invisible remains undetected. Wider deployment then increases the exposure of that unresolved deficiency.

Residual composition and consequence are therefore separate but connected. Residual selection asks what kinds of problems remain. Evaluation lag asks which problems are newly revealed or remain undetected as the system's reach grows.

The long-horizon example directly illustrates **capability outrunning an existing evaluation regime** and thereby revealing a class of problems that earlier evaluations missed. Residual concentration in hard-to-evaluate dimensions remains a separate, conditional claim that depends on the differential-resolution hypothesis in Section 3.

Capability can simultaneously improve monitoring, self-correction, and safety research. The cited laboratory report provides one documented instance of evaluation lag rather than evidence of a universal pattern.

The paper therefore treats the mechanism conditionally: when hard-to-evaluate dimensions also resolve more slowly, evaluation lag can help them persist while average performance improves and deployment expands.

The next source of movement lies outside the artificial system itself. The human reference used to interpret, demand, and evaluate AI performance can change as AI becomes embedded in human activity.

5. The Human Reference Is Not Fully Stationary

The **human reference** is the evolving set of human capabilities, practices, expectations, and evaluative standards against which AI performance is interpreted. The question here is whether changes in those reference conditions shape later AI development.

A long tradition of distributed and extended cognition treats tools and environments as potentially integral to cognitive activity rather than as passive accessories [12]. AI extends that familiar problem into a system that is adaptive, generative, and capable of participating across many stages of intellectual work.

5.1 Assisted performance and practiced capability

Research on cognitive offloading supports one important distinction: assisted performance is not the same as independent capability.

A 2026 review in *Trends in Cognitive Sciences* concluded that offloading cognition to AI can impede skill acquisition and contribute to skill decay in some settings, while emphasizing that outcomes depend strongly on how AI is used [13]. In a controlled logic-puzzle study, greater reliance on AI improved assisted performance but was associated with weaker subsequent unassisted performance, whereas independent reasoning during the assisted phase was associated with greater gains [14].

These studies establish a task-specific result: what people can do with AI and what they continue to practice or perform unaided can move differently.

That distinction matters because future AI systems are evaluated by humans whose practices can themselves change over time. Changes in evaluator, user, or domain-expert practice can alter the quality and nature of oversight.

5.2 Value, distribution, and feedback into development

Human capability can also move without any loss of underlying skill.

If AI makes routine drafting inexpensive, drafting skill may remain intact while its relative scarcity declines. Verification, judgment, problem formulation, accountability, or interpersonal trust may become more valuable. Cognitive work can also migrate into a coupled human–AI system rather than remaining concentrated in the individual.

These effects matter for later AI development through four concrete channels:

- **evaluation and oversight:** the capabilities of human evaluators affect what errors are detected and what behavior is rewarded;
- **training and feedback data:** human practices altered by AI can change the data later systems learn from;
- **demand and expectations:** widespread use changes what users and institutions ask later systems to do;
- **division of labor:** automation changes which human contributions remain scarce and therefore which bottlenecks matter.

These four channels provide the strongest basis for treating the human reference as partly endogenous to prior AI use.

AI increasingly operates against a human reference shaped by prior technology, institutions, and earlier AI use.

This movement also connects directly to successor development. If AI changes which human contributions are practiced, valuable, or available, it changes the human side of the process that builds the next generation of AI. The next question is therefore not only how AI changes human capability, but how AI itself enters the development process.

6. When AI Participates in AI Development

AI systems are increasingly used inside AI research and engineering. They write code, generate synthetic data, conduct evaluations, search literature, propose experiments, analyze results, and assist with research workflows.

Anthropic reported in 2026, using one internal measurement framework, that Claude “led” 26% of measured AI research and engineering work and collaborated substantially on more than 90%, while not operating fully autonomously in any measured subset [15]. The figure illustrates substantial participation within that measured environment; cross-lab automation levels remain an open empirical question.

The important development relation is therefore increasingly mixed:

$$\text{Human} + \text{AI}_t \rightarrow \text{AI}_{t+1}$$

Later systems can inherit substantive developmental contributions produced by earlier systems while humans may still retain substantial control over objectives, evaluation criteria, and deployment.

6.1 Variation is not selection

The distinction between **variation** and **selection** is central.

AI may increasingly generate candidate code, experiments, training procedures, architectures, or data without controlling what counts as improvement.

Selection itself can be decomposed into three functions:

1. **criterion setting** — deciding what counts as better;
2. **evaluation execution** — measuring candidates against those criteria;
3. **successor decision** — deciding which system or method actually advances.

AI can already perform parts of evaluation execution in benchmarked settings while humans or institutions retain substantial authority over criterion setting and successor decisions. Markets, users, regulation, infrastructure, and capital allocation also exert external selection pressures that are neither purely human-developer nor purely AI-generated.

This decomposition turns “AI selection” from a single threshold into several separable functions.

It also clarifies a possible trajectory: machine-generated variation may expand faster than machine authority over the criteria by which successors are chosen.

6.2 Machine-derived contribution can help or degrade development

The value of machine-generated developmental material depends on how it is produced, tested, and incorporated.

AI-run experiments, adversarial cases, externally verified search results, and discovered optimization strategies can add useful information. Recursive reuse of model-generated outputs can do the opposite. Shumailov et al. showed that indiscriminate recursive training on generated data can produce model collapse and loss of information about the original distribution [16].

The important distinction is therefore between machine-generated material that is independently tested or informative and material that mainly recycles earlier approximation error.

This supports the following developmental-lineage proposition:

Later AI systems can become increasingly machine-derived in their developmental inputs while human or institutional authority can remain substantial in reproduction, criterion setting, and successor selection.

6.3 Automation can move the human bottleneck

AI participation in AI R&D also provides a second application of residual selection.

As coding, experiment execution, search, and benchmark-driven optimization become easier to automate, remaining human work may become more concentrated in tasks such as problem choice, interpretation of ambiguous evidence, objective setting, governance, and tradeoff judgment.

Whether this concentration occurs is an empirical question. Some judgment-intensive tasks may also become highly automatable. **Does the human share of AI development become increasingly concentrated in weakly specified and judgment-intensive functions as more structured work is automated?**

That question links machine-derived development back to the paper’s central mechanism rather than treating it as a separate story.

It also creates the setting in which PAC becomes most useful: frontier developers must act on diagnoses that may remain only weakly identified while the system being diagnosed continues to change.

7. PAC, Identification, and a Moving Frontier

The preceding sections describe a development frontier that can change through differential resolution, evaluation lag, changes in the human reference, and increasing AI participation in R&D. The practical problem is to determine which explanation of the current limitation is sufficiently distinguishable to guide action.

At any frontier, several explanations can fit the same observed limitation. They become decision-relevant when they imply different things to try next and can be distinguished before the relevant structure changes again.

Prediction–Attribution Conjugacy (PAC) separates these two requirements [4].

Structural conjugacy exists when competing decision-relevant attributions imply different prospective outcomes under at least one common informative condition.

Identified conjugacy requires something stronger: feasible observations or interventions must provide enough diagnostic variation to distinguish those accounts in time for the decision.

That distinction is useful for frontier AI development because many plausible attributions are structurally different but weakly identified in practice.

A slowdown or persistent deficiency might be attributed to architecture, data, compute allocation, inference, evaluation, or a changing target. If those accounts imply different interventions, the relation can be represented compactly as:

$$A_i \rightarrow P_i \rightarrow D_i$$

where an attribution A_i generates a prospective implication P_i , which motivates an intervention D_i .

Chinchilla is a relatively favorable case. A compute-allocation attribution generated a concrete prospective test: under a fixed compute budget, a smaller model trained on substantially more data should outperform a larger undertrained one. The resulting experiment provided meaningful diagnostic evidence [2].

Other frontier questions are less identified. Large training runs bundle architecture, data, compute, post-training, tools, and evaluation changes. Even when an intervention succeeds, several explanations may remain compatible with the result.

This is especially important for the possibility of a future **approximation-regime change**.

A regime change means more than a new product name or an additional subsystem. Here it refers to a materially different approximation mechanism whose gains cannot be reproduced by reasonable variations within the existing broad mechanism under comparable constraints.

“Regime” is therefore a judgment category rather than a sharply observable boundary. What counts as “reasonable” must be judged against the development options actually available at the time, so the boundary may remain contestable even retrospectively.

An attribution that the current broad mechanism itself is binding would imply that a materially different mechanism should produce gains that continued within-regime adjustments increasingly cannot. But this is difficult to identify prospectively because compute, data, inference, and system composition remain available rival explanations.

PAC disciplines the question while remaining agnostic about which development path will prevail.

7.1 The identification race

The connection becomes stronger when the frontier itself moves while developers are trying to understand it.

The PAC paper describes an exploratory **identification race** using the conceptual ratio

$$\rho = \lambda_i / \lambda_r$$

where λ_i is the rate at which evidence reduces uncertainty among competing attributions and λ_r is the rate at which the relevant attributional structure changes [4].

If diagnostic learning substantially outruns structural change, understanding can stabilize.

If the two proceed on similar timescales—or structural change is faster—frontier diagnoses can remain provisional.

The rate of structural change varies across AI-development problems. Some relationships—such as loss–compute scaling within a stable training regime—may persist long enough for diagnostic learning to accumulate. Other parts of the frontier, including agentic behavior, deployment conditions, tool use, or evaluation regimes, can change more rapidly. The identification race should therefore be applied to the particular attributional structure being studied rather than to “AI development” as a single moving object.

PAC's specific role in the paper is to ask whether competing frontier explanations are not only structurally different but **identifiable before the relevant structure changes enough to make the old distinction obsolete**.

As AI research agents take on more of the development loop, the same issue can arise inside AI-assisted R&D: attribution, prediction, intervention, and system change may become increasingly compressed in time.

8. What Makes the Frontier Partly Endogenous

The argument can now be assembled around one development frontier and four mechanisms that reshape it.

The **development frontier** is the structure of unresolved problems that materially shape further AI advancement.

Its next state can be influenced by at least four internal mechanisms described here:

1. **differential resolution:** some problems are resolved faster than others, changing residual composition;
2. **evaluation lag:** new capability can reveal problems that earlier evaluation regimes did not detect;
3. **human-reference change:** prior AI use can change evaluation, data, expectations, and division of labor;
4. **AI-mediated development:** earlier AI can contribute directly to the production and evaluation of successor systems.

These mechanisms make the frontier **partly endogenous** to previous progress.

The qualifier **partly** matters because AI development also faces externally supplied constraints: chip supply, energy, capital, data access, regulation, institutional choices, market demand, and geopolitical conditions. Any of these can dominate at a particular time.

The mechanisms also generate observable conditions under which their explanatory reach should contract:

- prospectively coded dimensions with lower specifiability or evaluability do not show slower observed resolution over pre-specified comparison periods than otherwise comparable clearer dimensions;
- changes in human practices and expectations show little detectable effect on later evaluation, training or feedback data, demand, or oversight in settings where such feedback is expected;
- increasing AI participation in R&D does not shift the composition of remaining human work toward less structured or more judgment-intensive functions;
- evaluation systems routinely detect newly enabled failure classes before or at deployment, leaving little systematic lag between capability expansion and the discovery of important missed behaviors;
- or careful case analysis repeatedly finds that movement in the development frontier is better explained by externally supplied constraints than by the mechanisms proposed here.

The final condition is difficult to apportion cleanly because many “external” constraints—capital, demand, regulation, or infrastructure—can themselves respond to AI progress. Here they are treated as external to the paper’s technical mechanisms rather than as strictly exogenous in an economic sense.

The paper therefore offers a set of conditional mechanisms. Its testable content lies in those mechanisms rather than in the bare possibility that progress can change the frontier.

Its central claim is that successful AI progress can alter the composition and character of the next development problem. If particular channels prove weak, the explanatory reach of the account should contract accordingly.

9. Limits and Conclusion

This paper began from a simple observation: AI development can change the kind of problem that remains after each major advance. Bottleneck migration is familiar in technological development, cognition has long been distributed across humans and tools, and technological change has always altered the environment in which later innovation occurs [6], [12]. The contribution here is to place these mechanisms within a common account of frontier change and derive conditions under which successful AI development changes not only the scale but the composition, visibility, and attribution of the problems that remain.

Artificial systems can improve through mechanisms that differ from the human processes they approximate. If well-specified and easily evaluated problems resolve faster than weakly specified or hard-to-evaluate ones, the residual can change composition. Capability can then outrun evaluation, allowing unresolved deficiencies to persist into wider deployment. AI use can alter the human reference through changes in evaluation, data, expectations, and division of labor. And AI can increasingly participate in the research process that produces its successors.

Several future paths remain compatible with the argument. A new approximation regime may solve today’s difficult problems; external constraints may dominate later development; and some frontier attributions may remain weakly identified as the system changes. These possibilities define the scope of the argument.

The familiar question is:

How close can AI come to human intelligence?

This paper adds a different set of questions:

What kinds of development problems remain after each major advance? Which are resolved more slowly, which are newly revealed, how does the human reference change, and can we identify the next bottleneck before the frontier changes again?

Whether a fixed ceiling exists remains open. AI evolution can still be understood as an advancing approximation frontier whose composition is repeatedly altered by the progress already made.

The development problem is therefore not simply the same problem becoming smaller.

Under the mechanisms examined here, it can change in kind as well as scale.

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